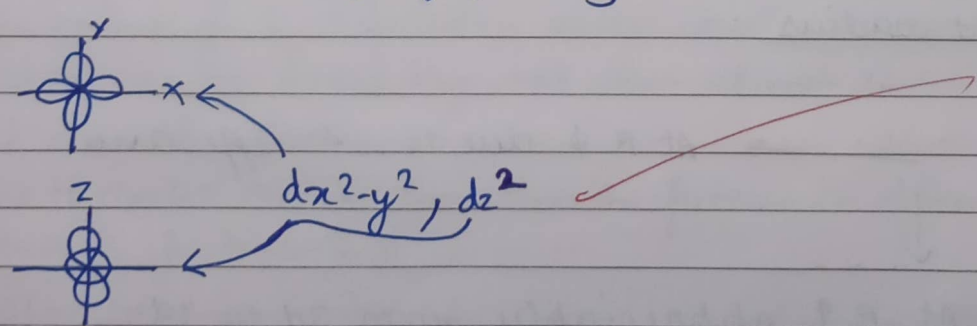
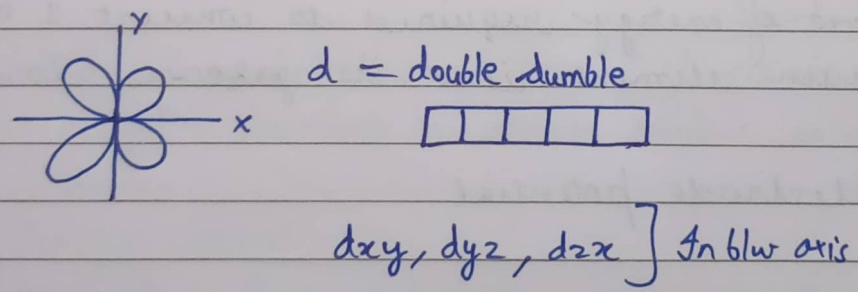
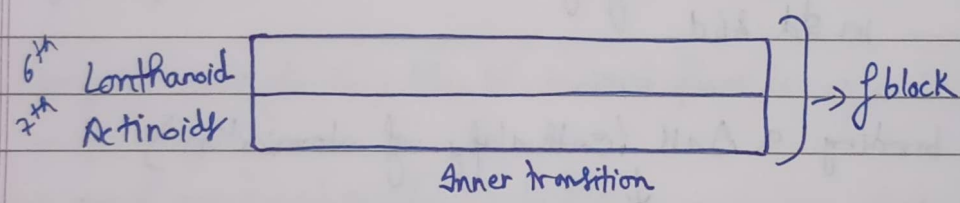
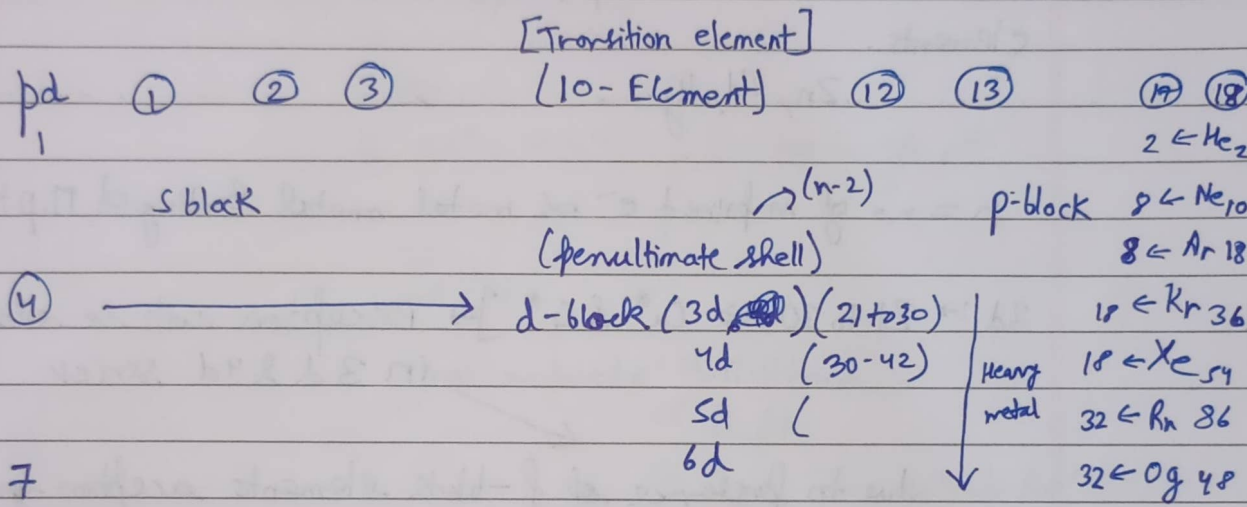


soluble
Monday

Ch-4 (Inorganic) D & F block element

class
work



d-block

4th pd 3d series $\rightarrow$

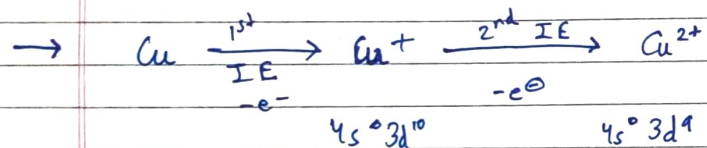
21	22	23	* 24	25	26	27	28	29	30
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
$[Ar] 4s^2 3d^1$	$[Ar] 4s^2 3d^2$	$[Ar] 4s^2 3d^3$	$[Ar] 4s^1 3d^5$	$[Ar] 4s^2 3d^5$	$[Ar] 4s^2 3d^6$	$[Ar] 4s^2 3d^7$	$[Ar] 4s^2 3d^8$	$[Ar] 4s^1 3d^{10}$	$[Ar] 4s^2 3d^{10}$
n = 1	2	3	6	5	4	3	2	1	0

10/8/26

Q Why transition metal show variable oxidation state?

- because due to comparable energy of ns & $(n-1)d$ orbitals
- Q Which transition metal do not show variable oxidation state?
- Sc^{3+} show only 1 oxidation state which is +3, Sc^{3+} because it gain Noble gas e^- configuration after loss of $3e^-$ (Cu^+)

Q Why Copper⁺ is unstable in Aq. medium which Cu^{2+} is stable?

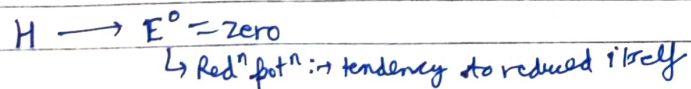
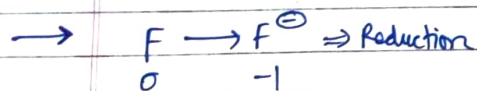


In aqueous medium hydration enthalpy play its role and show high value for Cu^{2+} due to its small size. High hydration enthalpy indicates more -ive value (exothermic)

High $\Delta_{hyd} H \Rightarrow$ -ive value [Exothermic]
E/q evolved
 $\hookrightarrow$ compensates 2^{nd} IE of Cu.

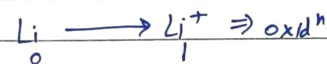
* In Aq medium Cu^+ show disproportionation reaction
 $Cu^+ \longrightarrow Cu + Cu^{2+}$

Q Copper metal does not evolved H_2 gas when we react with HCl and H_2SO_4 while in case of HNO_3 it will release H_2 ?



Strong

$LiAlH_4$



→ HNO_3 is the good oxidising agent because its reduction potential is more (+ive) ($E^{\ominus}$) in comparison to H_2 .

* Properties $\rightarrow$

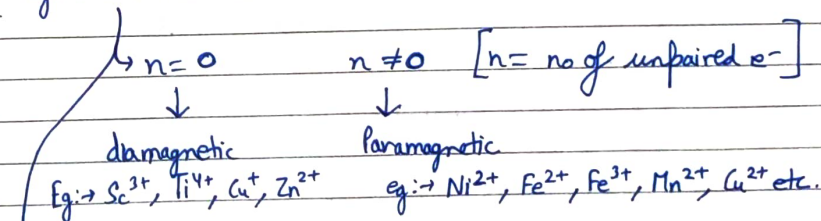
1. Complex formⁿ $\rightarrow$ small size
2. Coloured ion $\rightarrow$ number of unpaired e^-
3. Magnetic movement $\rightarrow \sqrt{n(n+2)} = \mu$, $n = \text{no of unpaired } e^-$
4. Interstitial compound $\rightarrow$ vacant d-orbitals
5. Catalytic behaviour $\rightarrow$ solid surface
6. Alloy formⁿ $\rightarrow$ due to almost similar radius
 $\hookrightarrow$ eg: $\rightarrow$ stainless steel.

11/8/26

Q Why atomic radius of 2nd and 3rd transition series elements are almost similar?

→ due to Lanthanoid Contraction.

① Magnetic behaviour $\rightarrow$



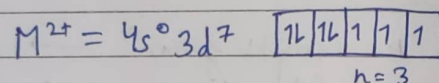
→ Calculation $\Rightarrow$ Magnetic moment (μ) = $\mu_s + \mu_o$
 $\hookrightarrow$ For d-orbital $\Rightarrow$ Quenching

So, $\mu = \mu_s$ (spin only)
 $= n \sqrt{n(n+2)} \therefore$ This value tell us about number of unpaired e^- .

Q Eg: $\rightarrow Mn^{2+}$, $n = ?$
 $\rightarrow 4s^0 3d^5$
 $n = 5$

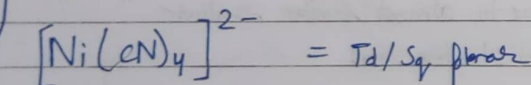
$$\mu = \sqrt{5(7)} = \sqrt{35} = 5.95 \text{ BM}$$

Q Calculate spin only moment for $M^{2+}(aq)$ ion $[Z=27]$?

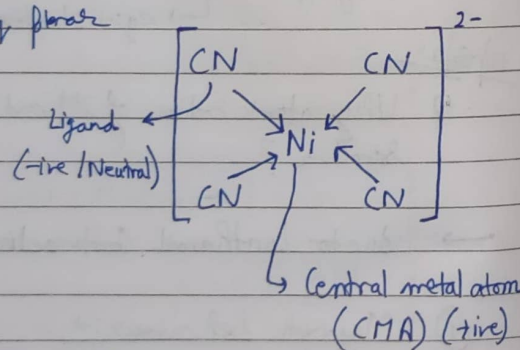


$$= \sqrt{n(n+2)} = \sqrt{3(5)} = \sqrt{15} = 3.9 \text{ BM}$$

② Complex formation: $\rightarrow$



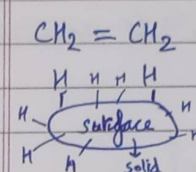
$x + 4(-1) = -2$
 $x = +2$



$\rightarrow$ Transition metal show variable Oxidⁿ state & form a small cation.

$\rightarrow$ CMA (TM) are having available d-orbital to form coordinate bond with Incoming Ligand (L)

③ Catalytic behaviour: $\rightarrow$



$\rightarrow$ provide solid surface to a particular substrate, to form temporary bond with it.
 $\rightarrow$ Temporary bond formation will take place only when the transition metal has available d-orbitals.
 $\rightarrow$ And result into, increase in rate of reacⁿ.

④ Interstitial compⁿ formⁿ

$\rightarrow$ Marked in book ① pointer

- Are those which are formed when small atom like H, C, N, B etc are trapped inside the crystal lattice of a pure metal. eg: $\rightarrow TiC, VH_{0.9}$ etc
- They will not follow any specific stoichiometry.
- Properties of Interstitial compounds:
 1. They are inert in nature
 2. Very high melting point
 3. Harder in comparison to pure metal
 4. They will show metallic substance.

~~⑤ Alloy formⁿ $\rightarrow$~~

⑤ Alloy formⁿ $\rightarrow$

$\rightarrow$ The metal which are combine together to form alloy should show difference in atomic radii with in 15% of each other.

$\rightarrow$ Transition metal do not show much variation of atomic radius. due to poor shielding of d orbital e⁻ eg: $\rightarrow$ Copper, Zinc = Brass

- Copper, Tin = Bronze
- Chromium, Vanadium, Tungsten, Molybdenum, Magnesium with Iron

$\downarrow$
Stainless steel

(b) Coloured ion formⁿ

Q Why Copper (Cu^+) is colourless while Cu^{2+} is coloured in aq. medium?

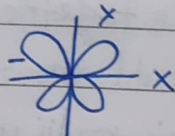
OR

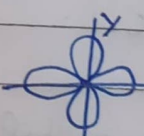
Q Why Sc^{3+} , Ti^{4+} , Cu^+ , Zn^{2+} are coloured less?

-
- no of unpaired e^- should be present = coloured.
 - unpaired e^- show d-d transition

↓
excitation of $d e^-$ to higher level ($t_{2g} \rightarrow e_g$)
↓
Absorb frequency comes under visible range.
↓
show complementary colour.

→ splitting of d-orbitals →

★ d_{xy}, d_{yz}, d_{zx}] = In b/w axis  x] Exist at angle other than 90°

$d_{x^2-y^2}, d_{z^2}$] = Along the axis  x] at 90° angle

